v) Time Travel & Data Recovery

A) What is Time Travel ?

B) Version Based Querying

C) Timestamp Based Querying

D) Restoring Deleted or Corrupted Data

Section 5 : Time Travel & Data Recovery

A) What is Delta Time Travel :

- Time Travel means the ability to read a table exactly as it existed at a previous point in time.

→ In Delta;
Every change create a new table version.

using SQL;

```
Select * From customers VERSION AS of 1
```

↓

Delta:
↳ reads txn logs up to version 1
↳ determines which files were active
↳ reads those parquet files.

(Result = exact snapshot of version 1)

B) Version Based Querying :

- we can query a table exactly as it existed at a specific version.
- This is called <u>version-Based Time Travel</u>

using Pyspark;

```
spark.read.format('delta')\
    .option("versionAsOf", 1)\
    .table("customers")
```

```
spark.read.format('delta')\
    .option("versionAsOf", 1)\
    .load("mnt/delta/customers")
```

viewing Table History:

```
DESCRIBE History customers
```

version	operation	timestamp
3	Delete	2024-01-10 10:30:00
⋮	⋮	⋮

c) Timestamp-Based Querying:

```
Select *
From customers
Timestamp As of '2024-01-10 10:30:00';
```

```
spark.read.format('delta')\
  .option("timestampAsOf", '2024-01-01 10:30:00')\
  .table('customers')
```

⇒ Find latest version ≤ timestamp

Timestamp = 10:06

versions available = 10:00, 10:05, 10:10

Delta chooses:

version 1 (10:05)

D) Restoring Deleted or Corrupted Data

Delta provides a Command;

```
Restore TABLE customers To version As of 5;
```

⇒ (This restores the table to the state it had in version 5)

```
Restore table table_name To version As of 5;
```

```
Restore table table_name
To Timestamp As of '_____';
```